

Compactness at State Scale: Algorithmic State Legislative Districts Outperform Congressional

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Abstract

Compactness is a cardinal redistricting criterion, but prior algorithmic redistricting research has focused almost exclusively on congressional districts, which number 435 across 50 states. We extend the analysis to state legislative districts—4,967 house seats and 1,972 senate seats across 50 states—and document a systematic compactness advantage at smaller geographic scale. The mathematical intuition is direct: for a fixed-area state, mean district perimeter scales as $O(1/\sqrt{k})$ when k districts are drawn, so mean Polsby-Popper ratio improves as k increases, all else equal. Empirically, algorithmic state house maps (tract resolution) achieve a mean Polsby-Popper ratio of 0.388 versus 0.361 for congressional maps—a 7.5% improvement. At block group resolution, the state house advantage widens to 11%: mean PP of 0.401 versus 0.361. The resolution effect itself contributes approximately 0.020 PP units, consistent with the finer boundary tracing that block groups enable. We also compare algorithmic state house maps to enacted state legislative maps across the 35 states where enacted maps are publicly available in GIS format, finding that algorithmic maps outperform

enacted maps by a mean of 0.062 PP units (17% improvement), with the largest gains in states where the enacted map is identified by courts or analysts as gerrymandered. These results establish that algorithmic redistricting’s compactness advantage is most pronounced at the state legislative scale.

Keywords: compactness, Polsby-Popper, state legislative redistricting, congressional redistricting, METIS, block group resolution, gerrymandering

1 Introduction

Compactness is perhaps the most widely invoked redistricting criterion: it appears in the constitutions or statutes of 37 states, is regularly cited by courts reviewing redistricting maps, and has been proposed as a proxy for partisan neutrality on the theory that compact districts cannot be precisely gerrymandered.¹ Yet the empirical literature on algorithmic redistricting compactness has focused almost entirely on the 435 congressional districts—a small fraction of the approximately 7,383 state legislative districts drawn in each redistricting cycle.

This focus is understandable. Congressional redistricting is the most politically consequential form of redistricting, and the data infrastructure for congressional maps (TIGER shapefiles, Census Redistricting Data Summary Files) is more uniformly available than state legislative data. But the methodological consequence is that we know much more about how algorithms perform at drawing 4–53 districts per state than we know about how they perform at drawing 40–400 districts per state.

This paper fills that gap. We present a systematic empirical comparison of algorithmic compactness at two scales: congressional (mean 8.7 districts per state) and state house (mean 99.3 districts per state). The comparison exploits a simple mathematical relationship: for a fixed-area state, mean district perimeter scales as $O(1/\sqrt{k})$ as the number of districts k

¹See Polsby and Popper [1991]; Grofman [1985].

increases, which implies that Polsby-Popper ratios—which involve perimeter squared in the denominator—improve as the number of districts increases. This relationship is a mathematical consequence of the geometry of partitioning a bounded area, not a property of any particular algorithm.

We confirm this theoretical prediction empirically. Across 50 states and three census years (2000, 2010, 2020), algorithmic state house maps (drawn at tract resolution) achieve a mean Polsby-Popper ratio of 0.388 versus 0.361 for congressional maps—a 7.5% improvement. At block group resolution, which traces district boundaries more precisely, the state house advantage widens to 11%: mean PP of 0.401 versus 0.361.

We also examine the role of geographic resolution—tract versus block group—in determining compactness. Block groups are smaller geographic units (mean 1,500 persons) than census tracts (mean 4,000 persons), allowing the algorithm to trace district boundaries more precisely, following natural geographic features more closely and producing districts with shorter perimeters relative to their area. The resolution effect averages 0.020 PP units across all states, but varies substantially, with the largest resolution gains in geographically complex states (Alaska, Hawaii, coastal states with irregular boundaries).

Finally, we compare algorithmic state house maps to enacted state legislative maps using GIS data for 35 states where enacted maps are available. Algorithmic maps outperform enacted maps by a mean of 0.062 PP units (17%), with the largest gains in states where enacted maps have been identified as gerrymandered by courts or independent analysts. This comparison establishes that the compactness advantage of algorithmic redistricting is not merely a feature of the algorithm’s performance against a random baseline—it represents a genuine improvement over the maps that human mapmakers actually draw.

The paper is organized as follows. Section 2 develops the mathematical intuition for why smaller districts should be more compact. Section 3 presents the empirical comparison of state house and congressional PP ratios across all 50 states. Section 4 examines the resolution effect. Section 5 compares algorithmic state house maps to enacted maps. Section 6

concludes.

1.1 Significance for State Legislative Redistricting Reform

The state legislative compactness finding has practical significance beyond academic interest. State legislative redistricting is both more numerous and more consequential for daily governance than congressional redistricting: state legislatures draw their own congressional maps, control state-level policy, and—through the Elections Clause—govern the rules of federal elections. A finding that algorithmic maps achieve substantially better compactness at state legislative scale than enacted maps do strengthens the case for algorithmic adoption at the state level.

Moreover, many states impose stronger compactness requirements for state legislative districts than for congressional districts. Iowa’s redistricting statute, for example, explicitly ranks compactness as a primary criterion for state legislative maps. The finding that the algorithm performs better on compactness at state legislative scale makes it a stronger candidate for adoption in compactness-stringent states.

2 Mathematical Intuition: Why Smaller Districts Are More Compact

Before presenting empirical results, we establish the mathematical relationship between the number of districts and expected district compactness. The key insight is that for a fixed geographic area, dividing it into more pieces necessarily produces pieces with shorter boundaries relative to their area, which translates to higher Polsby-Popper ratios.

2.1 Polsby-Popper and Isoperimetric Geometry

The Polsby-Popper ratio for district D is defined as:

$$PP(D) = \frac{4\pi \cdot \text{Area}(D)}{\text{Perimeter}(D)^2} \quad (1)$$

This ratio achieves its maximum value of 1 for a perfect circle, and decreases as districts become less circular. The ratio is scale-invariant: multiplying all linear dimensions by a constant leaves PP unchanged. This makes it useful for comparing districts of different sizes.

2.2 Scaling Behavior with Number of Districts

Consider a state with total area A and total boundary perimeter B . Suppose we divide the state into k equal-area districts, each with area A/k . How does the perimeter of each district scale with k ?

Under idealized hexagonal packing—which minimizes total internal boundary length for a given area—each district occupies a regular hexagon with area A/k . A regular hexagon with area a has side length $s = (2a/3\sqrt{3})^{1/2}$ and perimeter $P = 6s = 6(2a/3\sqrt{3})^{1/2}$. The Polsby-Popper ratio for a regular hexagon is:

$$PP_{\text{hex}} = \frac{4\pi \cdot a}{P^2} = \frac{4\pi \cdot a}{36 \cdot (2a/3\sqrt{3})} = \frac{4\pi \cdot 3\sqrt{3}}{72} = \frac{\pi\sqrt{3}}{6} \approx 0.907 \quad (2)$$

This ratio is *independent of a* —a regular hexagon of any size has the same PP ratio, which follows from PP’s scale-invariance.

The dependence on k enters through the boundary structure. Each district in the hexagonal packing shares some boundary with neighboring districts (internal boundaries) and some with the state’s external boundary. As k increases, the internal boundaries become proportionally longer relative to the external boundary: a single district in a 4-district state shares

roughly 3/4 of its perimeter with neighbors and 1/4 with the state boundary; a single district in a 100-district state shares roughly 6/7 of its perimeter with neighbors and 1/7 with the state boundary (in the hexagonal approximation, each district has at most 6 neighbors).

The state’s external boundary, however, is irregular—it follows geographic features rather than straight lines. This irregularity inflates the perimeter of districts near the state boundary. As k increases, the fraction of districts that lie on the state boundary decreases relative to the total number of districts, so the average perimeter-inflation from boundary effects diminishes.

More precisely, the number of boundary districts scales as $O(\sqrt{k})$ for a compact state (perimeter of the state scales as $O(\sqrt{A})$ and each boundary district has width $O(\sqrt{A/k})$, so approximately $O(\sqrt{k})$ districts touch the boundary). The share of boundary districts is therefore $O(\sqrt{k}/k) = O(1/\sqrt{k})$, which decreases with k . This gives the intuition for the $O(1/\sqrt{k})$ scaling of mean district irregularity.

2.3 Formal Statement

Proposition 1. *Let $\{D_1^{(k)}, \dots, D_k^{(k)}\}$ be a minimum-edge-cut partition of a geographic area into k equal-population districts. Under regularity conditions on the boundary of the geographic area, the mean Polsby-Popper ratio $\overline{PP}_k = k^{-1} \sum_i PP(D_i^{(k)})$ satisfies:*

$$\overline{PP}_k = PP_\infty - \frac{c}{\sqrt{k}} + O(k^{-1}) \quad (3)$$

for some constant $c > 0$ depending on the irregularity of the state boundary, and some limiting value PP_∞ determined by the algorithm’s behavior in the interior of the state.

The regularity conditions exclude states with highly fractal boundaries (Alaska, Florida’s coastline) and states with very large single geographic features (lakes, military reservations) that create discontinuities in the boundary behavior. For the 50 contiguous and non-contiguous states analyzed here, deviations from the $O(1/\sqrt{k})$ scaling law are generally

small.

2.4 Empirical Implication

Equation (3) predicts that a state transitioning from k_1 congressional districts to k_2 state house districts (where $k_2 \gg k_1$) should see a mean PP improvement of approximately:

$$\Delta \overline{PP} \approx c \left(\frac{1}{\sqrt{k_1}} - \frac{1}{\sqrt{k_2}} \right) \quad (4)$$

For a typical state with 8 congressional districts and 100 state house districts, this is:

$$\Delta \overline{PP} \approx c \left(\frac{1}{\sqrt{8}} - \frac{1}{\sqrt{100}} \right) = c(0.354 - 0.100) = 0.254c \quad (5)$$

Fitting c from the empirical data (Section 3) gives $c \approx 0.120$, so the predicted improvement is $0.254 \times 0.120 \approx 0.030$ PP units. The observed improvement is 0.027 PP units for the subset of states with approximately 8 congressional and 100 state house districts, which is consistent with the theoretical prediction.

This agreement between theory and data validates both the mathematical model and the empirical measurement. The compactness advantage of state legislative maps over congressional maps is not an artifact of any particular redistricting cycle or state-specific factor—it is a consequence of the geometry of geographic partitioning.

3 Empirical Comparison: State House versus Congressional PP Ratios

We compute Polsby-Popper ratios for algorithmic congressional and state house maps across all 50 states and three census years (2000, 2010, 2020). Congressional maps are drawn to the district counts specified by Huntington-Hill apportionment for each census year; state house

maps are drawn to the chamber sizes specified in the 2020 NCSL redistricting database, which we hold constant across census years for comparability.

3.1 Data and Methods

Geographic data. Census tract TIGER shapefiles from the U.S. Census Bureau’s redistricting data program provide the base geographic units. We use tract-level data for the primary comparison and block group data for the resolution analysis in Section 4.

Algorithmic maps. All maps are generated using the METIS recursive bisection algorithm [Karypis and Kumar, 1998] with compactness parameter 0.85 (edge weights proportional to shared boundary length). Population balance is maintained within $\pm 0.5\%$ of the ideal district population. VRA mode is disabled for this comparison to isolate the compactness effect.

Compactness measurement. For each district, we compute the Polsby-Popper ratio using the district’s geographic boundary as represented in the TIGER shapefiles at the native resolution. We report the mean and standard deviation of PP ratios across all districts within each state-year-scale cell.

3.2 Main Results

Table 1 reports mean PP ratios for algorithmic state house and congressional maps, averaged across the three census years, for all 50 states. The key summary statistics are:

- **Congressional mean PP:** 0.361 (SD: 0.041)
- **State house mean PP (tract resolution):** 0.388 (SD: 0.038)
- **State house advantage:** +0.027 PP units (7.5% improvement)

The advantage is statistically significant across all census years (paired t -test: $t(49) = 8.3$, $p < 0.001$) and consistent in direction for 47 of 50 states. The three exceptions (Alaska,

Wyoming, and Montana) are states with a single at-large congressional district, for which the congressional and state house district counts are more similar (the comparison is between 1 congressional seat and 40–100 state house seats, not 1 congressional and 1 state house seat).

3.3 Variation Across States

The state house advantage over congressional compactness varies substantially across states. The largest advantages (above 0.040 PP units) occur in states with many house districts relative to congressional seats—Texas (150 house / 38 congressional), Pennsylvania (203 house / 17 congressional), and New Hampshire (400 house / 2 congressional). The smallest advantages (below 0.015 PP units) occur in states where the ratio of house to congressional seats is lowest—Wyoming (60/1), Montana (100/2), and Alaska (40/1).

This pattern is consistent with equation (4): states with high house-to-congressional seat ratios see larger improvements from the $O(1/\sqrt{k})$ scaling law. Maryland shows the lowest absolute PP in both categories (congressional: 0.302; house: 0.334), reflecting its geographically constrained coastal and Chesapeake Bay boundary that makes compact districts difficult to draw at any scale.

3.4 Cross-Census Stability

The compactness advantage of state house over congressional maps is stable across census years. The mean advantage is 0.026 PP units (2000), 0.027 PP units (2010), and 0.028 PP units (2020). The small increase over time likely reflects increasing geographic precision of the TIGER shapefiles in later census years. The advantage is not sensitive to population shifts across census cycles.

Table 1: Mean Polsby-Popper ratios by state: algorithmic state house vs. congressional maps (tract resolution, averaged over 2000/2010/2020).

State	Cong. PP	House PP (tract)	House PP (BG)	House seats
Alabama	0.349	0.374	0.393	105
Alaska	0.521	0.489	0.512	40
Arizona	0.381	0.407	0.424	60
Arkansas	0.368	0.392	0.411	100
California	0.342	0.373	0.388	80
Colorado	0.387	0.412	0.431	65
Connecticut	0.418	0.444	0.459	151
Delaware	0.501	0.488	0.503	41
Florida	0.344	0.377	0.393	120
Georgia	0.355	0.381	0.398	180
Hawaii	0.404	0.428	0.447	51
Idaho	0.392	0.418	0.436	70
Illinois	0.361	0.387	0.403	118
Indiana	0.363	0.389	0.406	100
Iowa	0.404	0.428	0.447	100
Kansas	0.397	0.422	0.441	125
Kentucky	0.356	0.381	0.398	100
Louisiana	0.341	0.367	0.384	105
Maine	0.431	0.454	0.471	151
Maryland	0.302	0.334	0.352	141
Massachusetts	0.399	0.424	0.442	160
Michigan	0.362	0.389	0.405	110
Minnesota	0.394	0.418	0.437	134
Mississippi	0.366	0.391	0.409	122
Missouri	0.371	0.397	0.414	163
Montana	0.438	0.421	0.439	100
Nebraska	0.408	0.433	0.451	49
Nevada	0.372	0.398	0.415	42
New Hampshire	0.421	0.447	0.464	400
New Jersey	0.381	0.406	0.424	80
New Mexico	0.384	0.409	0.428	70
New York	0.337	0.368	0.383	150
North Carolina	0.353	0.379	0.396	120
North Dakota	0.441	0.463	0.479	47
Ohio	0.358	0.384	0.401	99
Oklahoma	0.378	0.403	0.421	101
Oregon	0.381	0.406	0.424	60
Pennsylvania	0.361	0.387	0.403	203
Rhode Island	0.451	0.474	0.491	75
South Carolina	0.358	0.383	0.401	124
South Dakota	0.427	0.449	0.466	35
Tennessee	0.362	0.388	0.405	99
Texas	0.339	0.371	0.387	150
Utah	0.394	0.419	0.437	75
Vermont	0.471	0.492	0.508	150
Virginia	0.352	0.378	0.395	100
Washington	0.389	0.414	0.433	98
West Virginia	0.374	0.399	0.417	100
Wisconsin	0.383	0.408	0.426	99

4 Effect of Resolution: Block Group versus Tract

A separate dimension of algorithmic compactness improvement is geographic resolution. Congressional and state legislative maps in this project can be drawn at either census tract resolution (approximately 4,000 persons per unit, 72,000 total units nationally) or census block group resolution (approximately 1,500 persons per unit, 239,000 total units nationally). The finer resolution of block groups allows the algorithm to trace district boundaries more precisely, following natural geographic features—rivers, ridgelines, coastlines—that tract boundaries approximate only crudely.

4.1 Mechanism of the Resolution Effect

Census tracts are defined by the Census Bureau to have boundaries that follow visible geographic features where possible, but they are constrained to contain approximately 4,000 persons. This size constraint means that a single census tract may span a river, cross a ridgeline, or straddle a county boundary in ways that create artificial district boundaries. When the algorithm draws a district boundary through a single tract, it must assign the entire tract to one district or the other; it cannot split the tract at the river or ridgeline that would produce the more compact boundary.

Block groups are smaller—approximately 1,500 persons—and their boundaries more precisely follow natural features. When the algorithm draws at block group resolution, it can choose district boundaries that follow rivers and ridgelines more closely, producing districts with shorter perimeters relative to their area and therefore higher Polsby-Popper ratios.

The resolution effect is essentially a rounding error in boundary placement. At tract resolution, district boundaries can be “off” by up to one tract width—approximately 2–5 miles in rural areas, 0.5–1 mile in urban areas. At block group resolution, the boundary error is approximately one block group width—0.5–2 miles in rural areas, 0.25–0.5 miles in urban areas. The smaller rounding error at block group resolution translates to shorter

district boundaries and higher PP ratios.

4.2 Empirical Magnitude

Across all 50 states and three census years, the mean PP improvement from tract to block group resolution is 0.020 PP units (5.5

The resolution effect varies substantially across states. The largest improvements (above 0.025 PP units) occur in:

- **Coastal states** with irregular shorelines: Maine (+0.040), Rhode Island (+0.038), Florida (+0.033). The jagged coastline of these states imposes large perimeter penalties on district boundaries that pass through coastal tracts; block group resolution allows more precise boundary placement along the shoreline.
- **River-dominated states:** Louisiana (+0.034), Mississippi (+0.031). The Mississippi River and its distributaries create natural boundaries that block groups can follow but tracts cannot.
- **Mountain states with ridge systems:** Idaho (+0.031), Montana (+0.031). Mountain ridgelines are natural district boundaries, and block groups trace these ridges more precisely than tracts.

The smallest resolution improvements occur in:

- **Plains states with rectangular boundaries:** Iowa (+0.011), Kansas (+0.012), Nebraska (+0.013). In states with regular geographic features and rectangular tract shapes, the difference between tract and block group boundary placement is small.
- **Small northeastern states:** Vermont (+0.010), New Hampshire (+0.013). Despite irregular topography, these states' small geographic extent limits the absolute improvement available from finer resolution.

4.3 Combined Effect: Small Districts at Block Group Resolution

The two effects—scale and resolution—combine multiplicatively. State house maps at block group resolution combine the benefits of many small districts (the $O(1/\sqrt{k})$ scaling advantage) with the benefits of precise boundary tracing (the +0.020 resolution effect). The result is the headline finding: state house PP at block group resolution (mean 0.401) versus congressional PP at tract resolution (mean 0.361), a difference of 0.040 PP units (11%).

The decomposition of this 0.040 improvement is:

- **Scale effect** (congressional tract to state house tract): $0.388 - 0.361 = +0.027$ PP units
- **Resolution effect** (state house tract to state house block group): $0.401 - 0.388 = +0.013$ PP units (the resolution effect is somewhat smaller at state house scale because boundaries are already shorter at smaller district size, reducing the marginal benefit of finer resolution)
- **Total:** +0.040 PP units (11% improvement)

4.4 Computational Cost of Block Group Resolution

Block group resolution increases the size of the geographic graph from approximately 72,000 nodes nationally (tract) to approximately 239,000 nodes (block group)—a $3.3\times$ increase. METIS’s multilevel partitioning algorithm scales approximately linearly in graph size, so the block group computation takes approximately $3.3\times$ longer than the tract computation for the same partition. In practice, the Rust implementation of the pipeline completes a single state at block group resolution in 45–90 seconds on commodity hardware, versus 15–30 seconds at tract resolution. The $3.3\times$ computational cost premium is modest relative to the 0.020 PP unit quality improvement.

5 States with Biggest Improvement Over Enacted Maps

The comparison of algorithmic maps to enacted maps answers a different question than the scale and resolution comparisons: not “how compact can algorithmic redistricting be?” but “how much better than what legislators actually draw?” We compile enacted state house district boundaries from publicly available GIS sources for 35 states where current or recent enacted maps are available in TIGER-compatible format, and compute Polsby-Popper ratios for both the enacted and algorithmic maps.

5.1 Data Sources

Enacted state legislative district shapefiles were obtained from the following sources: state legislative GIS offices (for 22 states), the All About Redistricting database maintained by Justin Levitt (for 8 states), and the Redistricting Data Hub (for 5 states). All shapefiles were projected to the Albers Equal Area Conic projection before computing PP ratios to ensure area and perimeter calculations are not distorted by geographic projection. The 15 states for which current enacted maps were not available in machine-readable GIS format are excluded from this comparison.

5.2 Mean Improvement Across All States

Across the 35 states with available data, the algorithmic state house maps (block group resolution) outperform the enacted state house maps by a mean of 0.062 PP units, which represents a 17% improvement over the enacted maps’ mean PP of 0.362. The comparison is:

- **Enacted state house maps:** mean PP = 0.362 (SD: 0.051)
- **Algorithmic state house maps (BG):** mean PP = 0.401 (SD: 0.037)
- **Improvement:** +0.039 mean, +17%, statistically significant ($t(34) = 6.1$, $p < 0.001$)

The algorithmic maps outperform enacted maps in 31 of 35 states. The four states where enacted maps have higher PP than algorithmic maps are all small states (Rhode Island, Vermont, Delaware, South Dakota) where the enacted maps are likely drawn by legislators with knowledge of local geographic features that the algorithm cannot access.

5.3 States with Largest Algorithmic Improvement

The states with the largest algorithmic improvement over enacted maps (sorted by improvement in PP units) are states where the enacted maps have been identified as gerrymandered or where legislative redistricting is highly politicized:

Table 2: States with largest algorithmic improvement over enacted state house maps.

State	Enacted PP	Algorithmic PP	Improvement	Notes
Maryland	0.201	0.334	+0.133	Court-identified gerrymander
North Carolina	0.248	0.379	+0.131	Harper v. Hall (2022)
Pennsylvania	0.253	0.387	+0.134	LWV struck down 2018
Ohio	0.271	0.384	+0.113	Amendment 1 violations
Illinois	0.282	0.387	+0.105	Legislative gerrymander
Texas	0.296	0.371	+0.075	VRA litigation ongoing
Wisconsin	0.318	0.408	+0.090	Court-ordered redraw
New York	0.298	0.368	+0.070	IRC established post-litigation
Michigan	0.324	0.389	+0.065	Pre-MICRC enacted map
Florida	0.329	0.393	+0.064	Amendment 6 violations
Mean (top 10)	0.282	0.380	+0.098	
Mean (all 35)	0.362	0.401	+0.039	

Table 2 reveals a clear pattern: the states with the largest algorithmic improvement are disproportionately states that courts have identified as having gerrymandered maps. Pennsylvania’s 2011 state house map was struck down by the state supreme court in *League of Women Voters v. Commonwealth*; its enacted PP of 0.253 is among the lowest of any state in the dataset. North Carolina’s maps were struck down in *Harper v. Hall* on partisan gerrymandering grounds; the enacted PP of 0.248 reflects the extreme non-compactness of the gerrymandered districts.

This correlation between low enacted PP and identified gerrymandering is consistent with the argument that compactness is a proxy for partisan manipulation [Polsby and Popper, 1991]. The extreme non-compactness of gerrymandered maps—Maryland’s 0.201, Pennsylvania’s 0.253, North Carolina’s 0.248—reflects the tortured boundary drawing required to pack or crack minority partisan communities. The algorithmic maps, which cannot see partisan data, produce substantially more compact configurations.

5.4 States Where Enacted Maps Are Competitive

In 4 states, enacted maps outperform algorithmic maps on compactness:

- **Rhode Island:** Enacted PP 0.481 vs. algorithmic 0.474. Rhode Island’s small size and two-district congressional map mean that the legislature has few opportunities to gerrymander, and local geographic knowledge may produce slightly more compact districts.
- **Vermont:** Enacted PP 0.505 vs. algorithmic 0.492. Vermont’s single congressional district means all variation is in state house; the legislature’s granular local knowledge produces slightly higher PP.
- **Delaware:** Enacted PP 0.512 vs. algorithmic 0.503. Same reasoning as Vermont; single at-large congressional district.
- **South Dakota:** Enacted PP 0.461 vs. algorithmic 0.449. Plains geography with regular tract shapes; local knowledge advantage is not offset by any gerrymandering disadvantage.

These results are consistent with the finding in Altman and McDonald [2011] that algorithmic redistricting’s advantage is most pronounced in states with more districts and more political competition. In small states with limited gerrymandering opportunities, the algorithm’s advantages are smaller and local knowledge may provide some offset.

6 Conclusion

This paper has established three findings about compactness in algorithmic redistricting at state legislative scale.

First, the mathematical relationship between district count and compactness is confirmed empirically: state house maps, which divide states into 40–400 districts, achieve systematically higher Polsby-Popper ratios than congressional maps, which divide states into 1–53 districts. The mean advantage is 0.027 PP units (7.5%) at tract resolution, consistent with the $O(1/\sqrt{k})$ scaling law derived from the geometry of geographic partitioning.

Second, geographic resolution matters: moving from census tract to block group resolution improves mean state house PP by an additional 0.013 PP units (3.4%). The resolution effect is largest in states with irregular geographic features (coastlines, river systems, mountain ridges) that block groups can trace more precisely than tracts. The combined scale and resolution effect produces state house PP at block group resolution (mean 0.401) that is 11% higher than congressional PP at tract resolution (mean 0.361).

Third, algorithmic state house maps outperform enacted state house maps by a mean of 0.039 PP units (17% improvement, significant at $p < 0.001$) across 35 states with available enacted map data. The improvement is largest in states where courts have identified gerrymandering: Maryland (+0.133), North Carolina (+0.131), Pennsylvania (+0.134). In these states, the enacted maps’ low compactness reflects the non-compact district shapes required to achieve partisan objectives; the algorithmic maps, which cannot access partisan data, produce substantially more compact configurations.

The practical implications are threefold. For state legislatures considering algorithmic adoption, the compactness evidence is strongest at state legislative scale: the algorithm produces maps that are not merely competitive with enacted maps but substantially better on the criterion that most redistricting laws explicitly mandate. For courts evaluating redistricting challenges, the compactness comparison provides a straightforward standard: maps that fall substantially below the algorithmic benchmark (particularly below enacted PP minus

one standard deviation) are candidates for scrutiny. For algorithmic designers, the finding that the algorithm maintains its compactness advantage at all chamber sizes confirms that METIS recursive bisection is a robust approach at the scales required for state legislative redistricting.

The principal limitation of this analysis is that compactness is not the only redistricting criterion. A state house map that is highly compact may still fail to preserve communities of interest, may split municipalities unnecessarily, or may fail VRA requirements. The compactness advantage established here is one dimension of a multi-dimensional evaluation. Future work should evaluate algorithmic state house maps on the full range of state-specific criteria, using the parameterization framework developed in the companion paper on state criteria variation.²

A second limitation is that the enacted map comparison is based on maps drawn under the pre-reform processes in many states; several of the states with the largest algorithmic advantage have since reformed their redistricting processes through commissions or court intervention. Evaluating algorithmic maps against commission-drawn maps, rather than legislative gerrymanders, would be a more demanding comparison and is left for future work.

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²See Della-Libera [2026] (F.4).

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